



NPS Homework 5

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KEYWORDS — statistics, nadaraya-watson

I. PART A

Consider the Nadaraya-Watson estimator

$$\hat{m}(x) = \frac{\sum_{j=1}^n K_h(x_j - x) y_j}{\sum_{i=1}^n K_h(x_i - x)}, \quad (1)$$

Where $(x_i, y_i), i = 1, \dots, n$, is a given sample. We note that the predicted value $\hat{y}_k$ is given by where H_k is row vector with j th entry given by $(H_k)_j = \frac{K_h(x_j - x_k)}{\sum_{i=1}^n K_h(x_i - x_k)}$. Hence the vector of predictions $\hat{y}$ is given by

$$\hat{y} = H y, \quad (2)$$

where $H_{kj} = (H_k)_j$.

II. PART B

In light of the calculations above, the generalized cross-validation measure of a Nadaraya Watson estimator with bandwidth h is given by

$$GCV(h) = \frac{1}{n} \sum_{i=1}^n \left(\frac{y_i - \hat{m}_h(x)}{1 - \text{Tr}(H)/n} \right)^2, \quad (3)$$

where the trace of the matrix H found above is

$$\text{Tr}(H) = \sum_{k=1}^n \frac{K_h(0)}{\sum_{i=1}^n K_h(x_i - x_k)}. \quad (4)$$

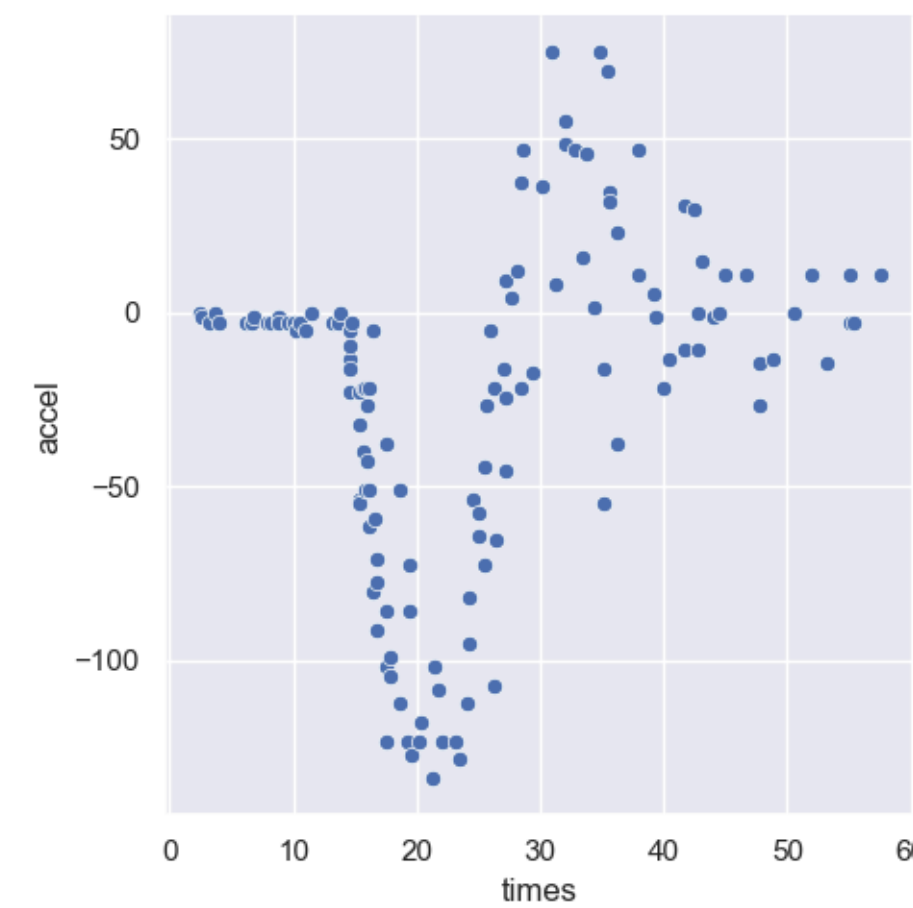
We thus use the GCV as a heuristic for determining the optimal bandwidth by seeking to minimize GCV over all (x_i, y_i) . Below we implement the GCV for a Nadaraya Watson estimator with standard normal kernel.

```
from scipy.stats import norm
import numpy as np
def K_h(x, h):
    return (1/h) * norm.pdf(x/h)

def trH(x, h):
    den = np.sum(K_h(x[None, :] - x[:, None], h), axis = 0)
    return np.sum( K_h(0, h) / den)
def GCV(x, y, m, h):
    return np.mean(((y - m) / (1 - trH(x, h)/x.shape[0]))**2)
```

We consider the “mcycle” dataset in the R Mass package shown below.

```
import pandas as pd
import seaborn as sns
import statsmodels.api as sm
cycle = sm.datasets.get_rdataset("mcycle", "MASS").data
sns.set_theme()
sns.relplot(cycle, x = "times", y = "accel")
```



We suppose that the acceleration variable y_i is related to the time variable x_i by some nonlinear relationship $y_i = m(x_i) + \epsilon$, where ϵ is some noise. Below we estimate $m(x)$ using 25 equispaced choices of bandwidth between $h = 0.1$ and $h = 5$, and compute the generalized cross-validation measure, defined above, for each, as well as the leave-one-out cross-validation metric defined by statsmodels. Based on the CV and GCV, the optimal bandwidth appears to be approximately $h \approx 1.53$.

```
from statsmodels.nonparametric.kernel_regression import KernelReg
import matplotlib.pyplot as plt

fig, plts = plt.subplots(nrows = 5, ncols = 5)
plts = plts.flatten()
for i, h in enumerate(np.linspace(0.1, 5, num = 25)):
    nw = KernelReg(cycle["accel"], cycle["times"], var_type="c", bw=[h])
    y_hat = np.array(nw.fit()[0])
    plts[i].scatter(cycle["times"], cycle["accel"], s = 1)
    plts[i].plot(cycle["times"], y_hat)
    cv = nw.cv_loo(bw = np.array([h]), func=nw.est["ll"])[0]
    gcv = GCV(cycle["times"].to_numpy(), cycle["accel"].to_numpy(), y_hat, h)
    plts[i].set_title(f"h = {h:.2f}, cv = {cv:.2f},\n gcv = {gcv:.2f}")

fig.set_size_inches(10, 10)
fig.tight_layout()
```

